Find the Promos

Why this problem

The brief states the bottleneck outright: an editor watches every episode by hand, and a new show ships every week. So the cost never stops, and unlike most things you'd point AI at, the answer is checkable. A timestamp either exists in the screenplay or it doesn't. Problem 1 is also the only one of the three that comes with real data, which meant I could show the tool working on Kuku's own material instead of on test cases I'd written to suit myself.

I scored all three against the five lenses before writing any code. Problem 1 won on quality checking and on the prototype for the same reason: a pick can be checked against its source, and the whole thing re-runs live on an episode it has never seen. Problem 2 would have meant answering legal questions with no legal background and nothing solid to ground the answers in. Problem 3 was the closer call. Reading the pack did turn up real defects, including inconsistent character labels and beats that run out of order, but rather than make those the submission I built the handling for them into the parser, which now reports 22 of them on every run.

The obvious objection is that Problem 1 is the obvious choice, so plenty of submissions will look alike. I decided to compete on method instead of topic.

Assumptions I had to make

A promo is a cut list, not a finished clip. The 20 to 60 second range and "hook inside 3 seconds" are hard limits in code, not preferences. Timestamps are relative to each episode's start, which I checked: all 50 begin at `00:00.00`, and there's no series-level clock anywhere in the pack. Splicing inside an episode is allowed, but only one promo crosses episodes, because stitching freely across 50 needs reliable character identity and this corpus has 64 speaker labels for 58 characters. The runtimes don't match the "1 to 2-minute" framing either: the median is 118.9 seconds, but 12 of the 50 run past 150 and the longest is 200.8. I went with the data.

And there's no ground truth for "good promo", so the scoring weights are mine, set by hand and written down in one table. Nothing here is learned, and I won't describe it that way.

The decision everything else follows from

Code owns time. The model owns judgment. The model never sees a timestamp and never writes one. It reads beats as `[b07] JULIANA (desperate): "..."` and answers with beat IDs, and then code looks each one up. A hallucinated timestamp isn't unlikely in this design. It's impossible.

That turned out to matter more than I expected. On episode 8, `qwen3.8-27b` chose a promo running `00:25.00` to `00:49.60` and then named as its hook a beat that starts 8.7 seconds *before* the promo begins. The description it wrote was fluent, and the moment it describes genuinely exists in that episode. It just isn't in the clip. Nobody catches that by reading the output, because it's perfectly consistent with itself. Only a check against the parsed beat table does, and two validators did.

How it works

Parse (code). 50 files become 3,727 timed beats. The parser reconciles labels against timestamps against beats and stops if they disagree, since a quiet partial parse would hand you promos built from half an episode.

Score (code). This is the part I got lucky on. The screenplays already label how almost every line is delivered: `(Cold, Commanding)`, `(Screaming, Desperate)`. 1,472 of 1,487 dialogue beats carry one, across 617 different values, so the tool never has to guess where the drama is. A lexicon I wrote by hand (90% hit rate) turns those tags into an intensity curve, and a promo is a shape on that curve: a spike in the opening seconds, a rising edge, and a cut while it's still high. From there it's 18,141 legal windows on real beat boundaries, 3,376 thrown out for having nothing in the first three seconds, and 171 shortlisted. All of it in 0.8 seconds, offline, free.

Select (model, one call per episode). Pick among the pre-checked windows and write the hook, the cliffhanger and one line on why. It also tags a hook type from DramaBox's published taxonomy, so an editor can spread types instead of posting four slaps in a row.

Validate (code). Six pass/fail checks, then near-duplicate suppression on the visual descriptions. 37 dropped, ten of them real cross-episode repeats, including episodes 31 and 32, which both end on a branding iron inches from the heroine's eye.

Gate (model, one call per promo). A fresh instance sees only the clip, with no logline, no episode number and nothing either side of it. It has to say who this is, what's at stake, what question it's left with, and whether it would keep watching.

The tool never posts anything. It turns "watch 50 episodes and scrub" into "look over 47 ranked, timestamped candidates", and the editor's keep and reject clicks get logged, which is what eventually replaces my opinion with real preferences.

How I know it's any good

Across all 92 promos, including rejected ones, `timestamps_exist` passes 92 of 92, and so do the other five checks. The blind gate tested 54, failed 7, left none untested, and scored ends-unresolved 1.00, character-identifiable 0.91, stakes-clear 0.76, would-keep-watching 0.65. Two runs give the same decision fingerprint, `c7b809ad92b976e7`. 42 of the 50 episodes end up with a promo.

None of that answers the real question: do the picks beat picking at random? So I tested it directly. Every pick goes up against a random legal window from the same episode: same length rules, same real beat boundaries, just not chosen. A blind judge sees two clips as plain text and never sees a score.

Every pair was judged twice, once with each clip shown first, because the judge turned out to prefer whichever clip it read second. Running both orders cancels that out instead of needing a correction afterwards. I counted by pair rather than by judgement, since two judgements of the same two clips aren't independent. Against any legal window, the tool was preferred in **both** orders on 24 of 38 pairs and the random window on 7, with 7 flipping. That's **77.4% of the 31 pairs that answered consistently, p=0.003**. Against a window that already passes the hook filter it's 22 to 8 with 8 flips, so **73.3%, p=0.016**. Chance is 50%.

That second comparison is the more useful number. Demanding a strong opening does a lot of the work by itself, and the tension scoring and the model's choice add a real edge on top of it rather than being the whole story. I'd rather report that split than the flattering headline.

Splicing turned out to cost comprehension. Continuous promos pass the gate at 0.886, spliced ones at 0.800, so the penalty I'd charged for the cut on instinct now has something behind it. Across two model families, agreement is only 0.50 IoU, with no overlap on 8 of 50 episodes and hook-type agreement 0.48. So a real part of this ranking is model taste rather than signal, which is an argument for keeping the editor in the loop.

I deliberately didn't build a hand-picked reference set as the headline measure. It asks whether the tool matched one person's taste, which isn't the job, and keeping one per show would recreate exactly the manual work the tool is meant to remove.

Measuring before building, three times over

This is the part I'd most want to be asked about, because in each case the obvious fix would have shipped looking fine and done nothing.

The duplicate check started life with a threshold of 0.45 on dialogue overlap. Before trusting it I measured all 1,225 promo pairs, and dialogue overlap peaks at 0.177, so it could never have fired. Repetition in this show is visual rather than verbal: the same image with different words. Switching to the visual descriptions put episodes 31 and 32 at 0.191, fourth highest of 1,225, and six of the top seven pairs turned out to be one or two episodes apart.

Then the model wrote "blood and dust flying" for a beat that says "stone dust", and blood appears nowhere in that episode. The fix I had in mind was matching description words against the clip, so I ran it first. It flagged 46 of 47 promos, because the frequent offenders are words like `declares` and `hanging`, which are paraphrase rather than invention. Narrowed to concrete physical nouns it's 2 of 47. The real rate is about one in fifty, and the obvious verifier would have fired on everything while catching nothing.

Then I noticed that `BLOND NOBLEMAN` speaks two beats from that same hook, and blond is one character away from blood. A corrupted character label would be far more alarming than embellishment, and it fits the evidence. So I tested it: same prompt, same model, speaker renamed so the string blond appears nowhere. The fabrication survived. It's cliché completion, not corruption. The more interesting explanation was the wrong one, and I'd have had no evidence for it.

Scale and cost

Measured rather than projected: one 50-episode show needs 107 model calls and about 175,000 tokens. Building it cost 127; probes and retries aren't free. At a show a week that's 2,600 episodes, roughly 5,600 calls and 9M tokens a year. The free tier covers it outright, and at Gemini Flash-Lite rates it'd be about $2 a year. Tokens are not the scaling cost and I'd say so in a planning meeting. What limits the free tier is 8,000 tokens a minute, which stretches a show to about 20 minutes of wall clock and disappears the moment someone pays. What costs real money is editor attention, and that's why the gate and the duplicate suppression exist. Provider is a flag, not a code change: Groq and Gemini are both verified end to end on different request schemas.

Where it falls over

It can't see. Shot quality, performance and pacing are invisible when your input is text, and the script tells me what gets said, not whether it lands. "Would keep watching" is the weakest gate criterion at 0.65, so clips are followable but not always compelling, and that gap is taste I can't measure. Eight episodes ship nothing at all, having lost their best candidate to the gate or to a duplicate. And the model's prose embellishes even though its timestamps can't, which I've scoped honestly rather than papered over: the timestamps are verified, the descriptions aren't.

One more, and it's the one I'd lead with in conversation. I built reveal detection, fixed the action lexicon so that 21 unmistakably violent beats stopped scoring zero, and forced the hook into the first second, which cut dead air at the front of promos from 2.4 seconds to 0.7. It changed 8 of 43 picks and surfaced a promo I'd hand-picked myself. I didn't ship it, because it invalidated the A/B result and I couldn't re-measure 76 blind judgements inside the free tier's rate limit before the deadline. Shipping better promos next to a headline number that described the old ones would have been the one dishonest thing in this submission. It's in the git history.

AI tools, and where they got it wrong

Claude for code and drafting, Groq-hosted `openai/gpt-oss-120b` as the pipeline model, `qwen/qwen3.8-27b` as the comparison, and Gemini to verify the provider swap. Five examples with before and after are in `docs/ai_mistakes_log.md`.

The one I learned most from was in my own evaluation code. When splicing started shipping, agreement with my reference spans jumped from 0.46 to 0.519, which looked like the feature had worked. It hadn't: my IoU measured a promo's outer envelope, so a two-piece promo got credit for the seconds the splice removes. The real segment union put it back to 0.465. Nothing errored, no test failed, and I only caught it because a feature I expected to be marginal produced a 13% gain and that didn't smell right. A metric that improves the moment you add code deserves more suspicion than one that doesn't.